

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	05 July 2026
Team ID	Solo Project – Individual Capstone
Project Name	Library Management System (ServiceNow)
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Role & User Management	Create Student role Create Librarian role Assign roles to individual users
FR-2	Book Catalog Management	Add/update Book records (Title, Author, ISBN) Track Status: Available, Issued, Lost (default = Available)
FR-3	Borrow Request Management	Student submits a Borrow Request (Requested By, Book, Request Date) Reference Qualifier restricts Book field to Status = Available
FR-4	Approval Workflow Automation	Flow Designer flow triggers on Status = Requested Route to Librarian for approval/rejection Auto-update Book status to Issued on approval Send email notification to requester
FR-5	Access Control	ACLs restrict Book table Write/Delete to Librarian; Read for Student & Librarian ACLs restrict Borrow Request Create to Student; Write/Delete to Librarian
FR-6	Reporting	“Most Borrowed Books” report grouped by Book, counted from approved requests Shared with Student and Librarian roles

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple Service Portal forms for both roles; UI Policy keeps forms clean by making Return Date mandatory only when a book is Issued.
NFR-2	Security	Role-based ACLs restrict Create/Write/Delete by role; students can only view/create their own Borrow Requests.
NFR-3	Reliability	Flow Designer automates status updates on every approval, removing manual data-entry errors between Book and Borrow Request records.
NFR-4	Performance	Reference Qualifier limits the Book picklist to Available records only, keeping request forms fast as the catalog grows.

NFR-5	Availability	Hosted on a ServiceNow cloud-based Personal Developer Instance (PDI), available per standard ServiceNow platform uptime.
NFR-6	Scalability	Modular table design (u_book, u_borrow_request) allows new related services (e.g. lab equipment requests, room booking) to be added without re-architecture.